

Dupire PINN — deep self-consistent local volatility

A forward-looking density from one day of option quotes

Wang Z., Shaa A., Privault N., Guet C.

One calibration, no historical prices, no forecasting model

7 August 2001 DAX — 217 call quotes, 5 maturities

The question

- From a **single day** of the DAX option market — 7 August 2001, 217 active call quotes across 5 maturities ($\tau = 0.123, 0.200, 0.373, 0.603, 0.871$ yr), spot $S_0 = 5,752.51$ — recover the full **probability density of the index at any future date**.
- No historical prices, no time-series model, no forecast. The only inputs are today's quotes and the requirement that the surface admit **no arbitrage**.
- The density follows from that requirement alone: if $C(T, K)$ is arbitrage-free, then by Breeden–Litzenberger

$$f_\tau(K) = e^{r\tau} \frac{\partial^2 C}{\partial K^2}(\tau, K)$$

is the risk-neutral density of S_τ — for **every** τ the surface is defined at, not only the 5 that happened to be quoted.

- **Three things must then be shown:** the second derivative is extracted correctly; the density is consistent with the model's own dynamics; and it stands up against what the index actually did.

The mechanism, in one slide

1. **Calibrate.** A network carries the normalised call price $\tilde{\varphi}(\tilde{t}, \tilde{k})$ through the **exp_k ansatz**

$$\tilde{\varphi} = \exp(-\tilde{k} \text{softplus}(a + \tilde{k}H)), \quad a = y + \log(1 - e^{-y}), \quad y = K_{\max}/S_0,$$

H the raw output. This makes $C(T, 0) = S_0$ and unit mass hold **by construction**, not as a fitted property — the boundary is not something the optimiser can trade away.

2. **Close the loop.** Training is self-consistent: the same network supplies the Dupire local volatility $\sigma_{\text{loc}}^2 = 2(\partial_T C + rK\partial_K C)/(K^2\partial_K^2 C)$, and the price surface must satisfy the Dupire PDE it implies.
3. **Differentiate.** $\partial^2 C/\partial K^2$ comes from nested automatic differentiation of the calibrated surface — analytic in the network, evaluated at any (τ, K) , with no interpolation between quoted expiries.

Rate $r = 4\%$; $K_{\max}$, $T_{\max}$ derived from the quotes, not set by hand.

One calibration $\rightarrow$ a density at every horizon

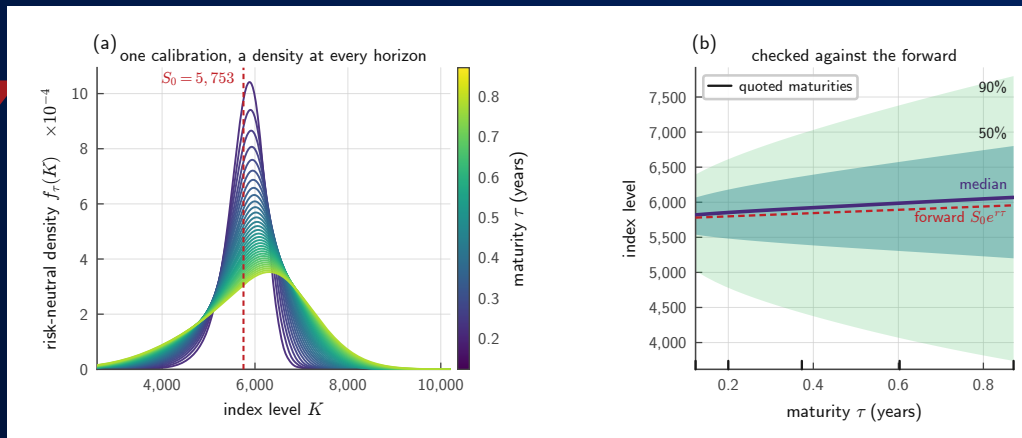


Figure 1. The risk-neutral density of the DAX implied by the 7 August 2001 calibration, shown (a) at 33 maturities spanning the quoted range $\tau \in [0.123, 0.871]$ yr, coloured by τ , and (b) as the corresponding quantile fan against the forward $S_0 e^{r\tau}$. The intermediate curves are not interpolated between fitted densities: each is read directly off the one calibrated surface, which is what makes a forward-looking distribution available at horizons that were never quoted. The distribution widens with τ and carries a heavier left tail than right — the equity-index skew, recovered without being imposed. **Caveat:** the axis stops at the longest quoted expiry; nothing is extrapolated in τ .

Why not just difference the quotes?

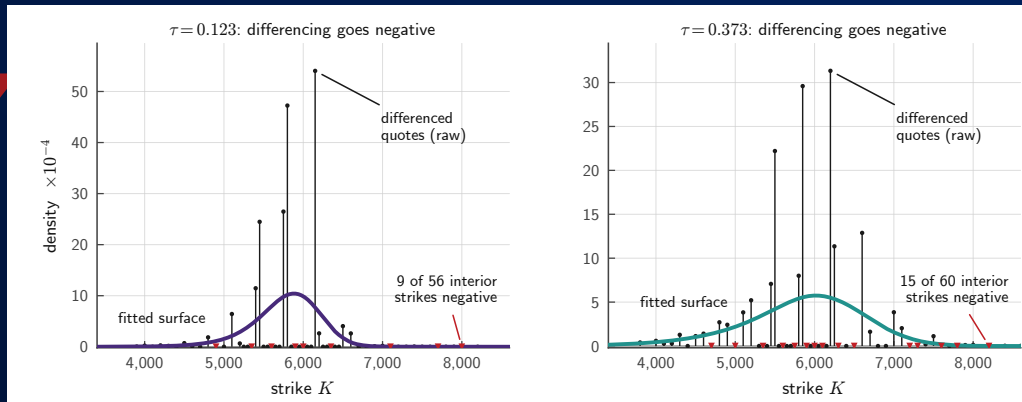


Figure 2. Breeden–Litzenberger applied *directly* to the quoted call prices, with no surface in between: a three-point second difference on the quoted strike ladder (stems), against the density from the fitted surface (curve), at the two maturities with the most quotes. The ladder is spaced 50–400 index points and the prices are rounded to the nearest 0.01, so the second difference is dominated by where the rounding happens to put a kink: it spikes to $2.7\text{--}6.9\times$ the true peak, collapses to zero between spikes, and goes **negative** at 36 of 207 interior strikes (6%–25% per maturity) — a negative probability density. This is why a smooth arbitrage-free surface must be fitted first, and why the finite-difference check on the next slide is applied to that surface rather than to these prices.

Is the second derivative right?

Two independent routes to the same second derivative agree to 2.2×10^{-3} of peak at worst

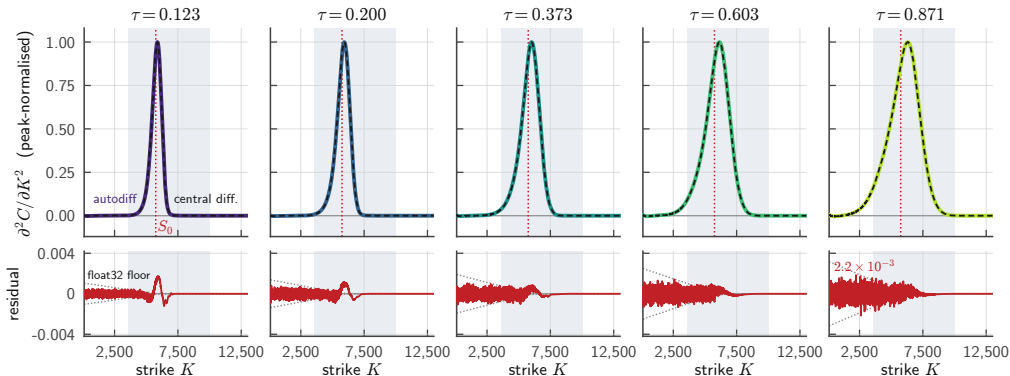


Figure 3. $\partial^2 C / \partial K^2$ at each quoted maturity, read two independent ways, both applied to the fitted surface, not to the quoted prices: the pipeline's nested automatic differentiation (colour) and a three-point central difference of the network's own call output at $K \pm 50$ (black dashed), each normalised by its own peak; the strip below shows the residual against the float32 stencil floor $4\varepsilon_{32}|C|/h^2$ (dotted). The two agree to 2.2×10^{-3} of peak at worst, and away from the money the residual is that floor — no numerical headroom remains. The excursion at the money exceeds it by $105\times$ and is genuine $h^2 C'''' / 12$ truncation. **Caveat:** both routes read the same weights, so this tests the extraction, not the calibration; outside the shaded trained range both extrapolate.

The step size is chosen, not guessed

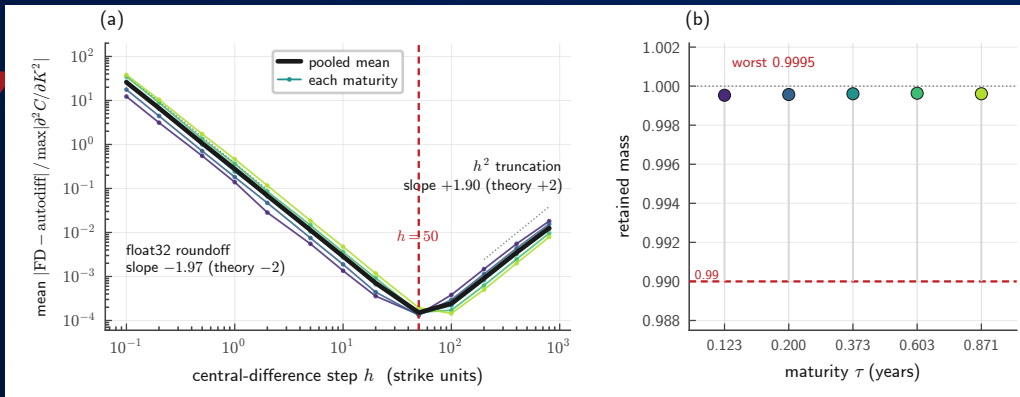


Figure 4. (a) Mean disagreement between the finite-difference and automatic derivatives against the step h , per maturity and pooled. The curve is the classical U: the left branch has fitted log-log slope -1.97 against the -2 predicted by float32 round-off, the right branch $+1.90$ against the $+2$ predicted by h^2 truncation. Recovering both exponents from a trained network is what licenses reading the minimum at $h = 50$ as numerically optimal rather than tuned, and confirms the residual in Figure 3 is numerical, not structural. (b) Mass retained on $K \in [100, 15, 000]$, worst 0.9995; the density mean reproduces the forward to 0.027%. **Caveat:** the density is mildly negative on up to 7.2% of grid points, all in the extrapolated tail outside the quoted strikes (worst excursion 0.19% of peak) — a convexity violation where there are no quotes.

Against what the index actually did

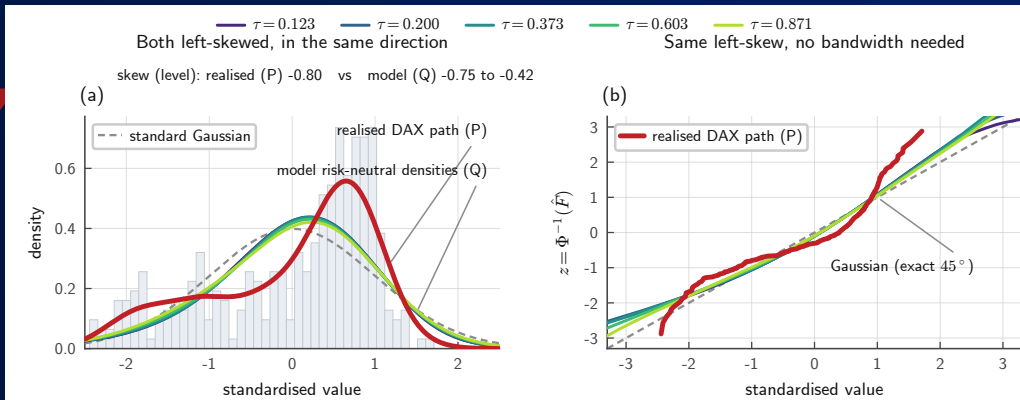


Figure 5. The realised DAX path over the 316 trading days from 2001-08-07 to 2002-11-01 (red), discounted and standardised exactly as in N. Privault's script, against the model's risk-neutral densities at the 5 quoted maturities (colour): (a) densities at a common bandwidth $h = 0.316$ taken from the coarser series, each variance-corrected to s.d. 1 so widths are comparable; (b) the bandwidth-free normal-quantile view, where a Gaussian is exactly the 45° line and curvature is skew. Both are clearly left-skewed and depart from the Gaussian in the same direction: realised skew -0.80 against -0.75 to -0.42 . This compares a risk-neutral (Q) density against one realised path (P) — different objects under different measures, $n = 1$ on the $\mathbb{P}$ side, no change of measure applied — so shape agreement is *suggestive, not a formal test*. Second caveat: the same comparison in log space is not usable, since moving the left grid edge between two windows that both retain ≥ 0.9995 of the mass shifts the log skew by 2.61 but the level skew by only 0.05 ($54\times$), so the log statistic reports where the grid was cut.

Two independent routes to the same density

Two independent routes to the density agree: $KS \leq 0.0047$ at every maturity

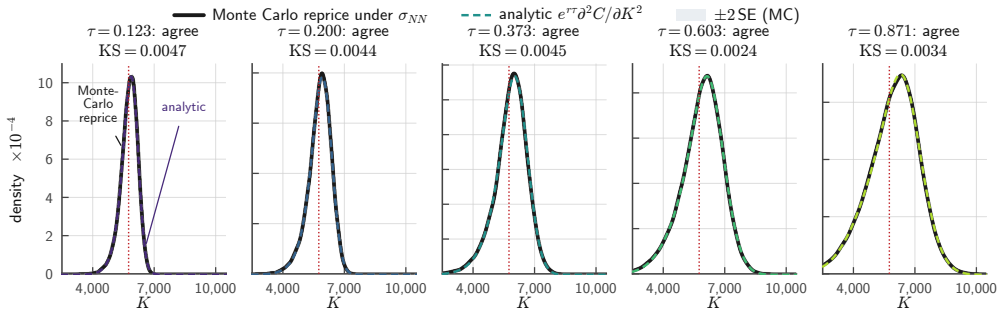


Figure 6. The analytic density $e^{r\tau} \partial^2 C / \partial K^2$ (dashed) against a Monte Carlo reprice of the *same* surface under its own local volatility σ_{NN} (black; 40,000 antithetic paths, 871 Euler steps), with the ± 2 MC standard-error band shaded. The two are obtained by genuinely different operations — one differentiates the price surface twice in strike, the other simulates the SDE that surface implies and never sees the prices — so agreement is an internal-consistency check, not a restatement. They agree to a Kolmogorov–Smirnov distance of 0.0047 at worst across all 5 maturities, and the simulated forward matches $S_0 e^{r\tau}$ to 0.022%. **Caveat:** at the shortest maturity the analytic density is 2.5% wider than the reprice, and refining the Euler step $8\times$ closes only 35% of that gap, so part is time discretisation and part a real disagreement at short τ .

The claim

- From **one day** of quoted option prices — 217 calls on 7 August 2001 — we recover a **full forward-looking probability density** of the DAX at any horizon out to the longest quoted expiry, using no historical prices and no forecasting model.
- It follows from no-arbitrage structure alone, via a calibrated Dupire surface whose boundary behaviour is built into the ansatz rather than fitted.
- It survives the checks that matter: extraction correct to 2.2×10^{-3} of peak, with both theoretical error exponents recovered ($-1.97, +1.90$); self-consistent with its own dynamics to $KS = 0.0047$; and its shape matches what the index did, in the one sense a single realised path can support.
- **What it is not:** a $\mathbb{P}$ -measure forecast. It is the market's risk-neutral view on 7 August 2001 — exactly what is wanted for pricing and hedging, and what must be adjusted for a risk premium before being read as a prediction.

Appendix — where the two densities differ

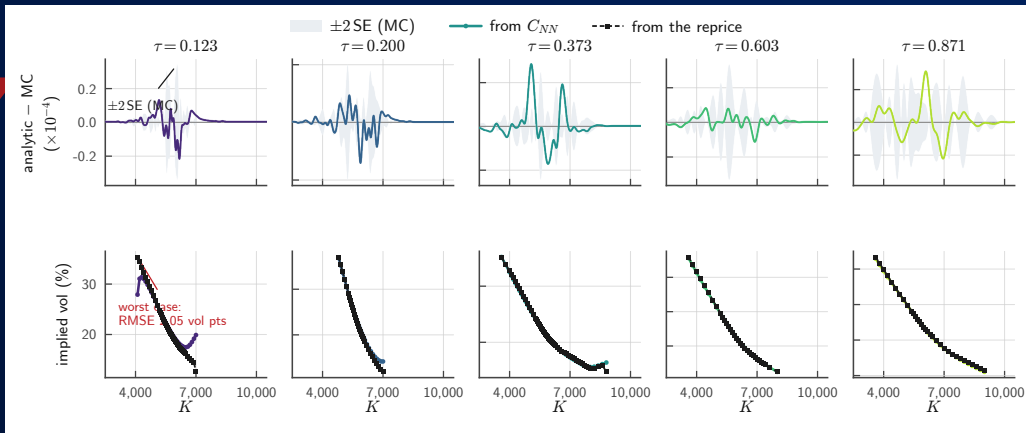


Figure A1. Companion to Figure 6. Top: the analytic-minus-reprice density residual against the ± 2 MC standard-error band. Bottom: implied volatility on the quoted strikes, from the network's call prices and from the reprice. The residual is structured rather than noisy — up to 51% of the grid lies outside twice the MC error at the shortest maturity — but small in the terms that matter for pricing: the implied-volatility disagreement is 2.05 vol points at the shortest maturity and at most 0.33 at every other one. The shortest maturity is where the Euler discretisation is coarsest relative to τ , consistent with the dt refinement reported in Figure 6.